Predicting Bilingual Proficiency: Eye-Tracking Analysis of Chinese-English Code-Switching Using Machine Learning

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Abstract

This study explores how machine learning (ML) analysis of eye-tracking data can reveal the relationship between code-switching and visual attention in bilinguals. Code-switching, the switching of languages during communication, offers insight into the cognitive mechanisms underlying bilingual processing. By analyzing how bilinguals process code-switched Chinese-English sentences, our research uses a Random Forests classification model to predict second language proficiency based on eye movement. This approach combines eye-tracking measurements with pattern recognition to identify how attention patterns during language switching correlate with proficiency levels. Our interdisciplinary methodology merges cognitive science, linguistics, and computational modeling to reshape how we understand bilingual language control. Predicting proficiency using real-time data enables more efficient language assessment tools for educational and clinical settings. Insights into cognitive load during language-switching can inform personalized teaching strategies, while this research also advances AI systems for bilingual language use, improving machine translation, natural language processing, and human-computer interactions. By examining the relationship between visual attention and language proficiency, the study aims to uncover how humans navigate multiple language systems, and has far-reaching implications in an increasingly multilingual world.

Methodology:

This study consists of two parts: leveraging existing eye-tracking data to develop an initial Random Forests classifier model and collecting our own eye-tracking data to refine the model.

1. Existing Data

We will start by using an open-source eye-tracking dataset from Chinese-English bilinguals, with 80 participants categorized by their English proficiency (low and high). The dataset includes key eye-tracking metrics such as Regression Path Duration, First Fixation Duration, Gaze Duration, and Regression In/Out Counts, providing insights into visual attention during code-switched sentence reading. This dataset will serve as the foundation for training our machine learning model.

2. Data Preprocessing and Feature Engineering

Data cleaning will involve filtering missing values, removing outliers, and performing quality checks. We will then use feature engineering to select the most relevant variables through correlation analysis and linear regression, ensuring the model focuses on the strongest predictors of bilingual proficiency.

3. Model Training

We will split the data into 70% training and 30% testing subsets. The model will be trained using the Random Forest classification technique, predicting language proficiency based on eye-tracking metrics. Model performance will be evaluated by classification accuracy.

4. Future Data Collection

To refine the model, we plan to collect our own eye-tracking data from bilingual participants and design our own linguistic stimuli for participants to use while collecting the data. This will allow us to tailor the data to our research needs, address gaps in the existing dataset, and improve the model's accuracy and generalizability.

5. Expected Outcomes

The outcome will be a machine learning model capable of predicting bilingual proficiency based on eye-tracking data, contributing to understanding cognitive processes in bilinguals and potentially improving language proficiency assessment tools in both educational and clinical contexts.

Paper 1: ChineseEEG: A Chinese Linguistic Corpora EEG Dataset for Semantic Alignment and Neural Decoding

Link

N=10

Experimental procedures

106 Participants were instructed to sit in an adjustable chair, whose eyes were approximately 67 cm away from the monitor (Dell, 107 width: 54 cm, height: 30.375 cm, resolution: 1,920×1,080 pixels, vertical refresh rate: 60 Hz), see Figure 1b.

They were tasked with reading a novel and were required to **keep their heads still and keep their gaze on the highlighted (red) Chinese characters moving across the screen, reading at a pace set by the program.** They were required to read an entire novel in multiple runs within a single session.

Each run is divided into **two** phases: **the eye-tracker calibration phase** and the **reading phase**, with a **break** between two adjacent runs to allow the experimenter to check the electrodes' impedance and add saline if necessary.

Each run includes either 3 to 4 chapters of The Little Prince or a single chapter of Garnett Dream, lasting approximately 30 minutes.

- For our experiment, we can use one sentence or phrases (10 min session)

The presentation of stimuli was managed using **PsychoPy v2023.2.327**, with the EGI PyNetstation v1.0.1 (this is for EEG so we may not need) module facilitating the connection between PsychoPy and EGI Netstation.

We also utilized **g3pylib package** to control our **eye-tracker to follow the eye movement trajectories of the participants.**

Phase 1: Eye-tracker calibration phase

At the beginning of each run, participants were required to undergo an eye-tracker calibration process. Initially, the message "Hello! Please press the spacebar to start

calibration" was displayed at the screen's center. Participants were instructed to keep their gaze at a fixation point, which sequentially appeared at the four corners and the center of the screen, each for 5 seconds. If the calibration failed, participants were prompted to start another calibration. Upon successful calibration, the message "Calibration successful! The page will automatically redirect in 5 seconds" was displayed at the center of the screen

Phase 2: Reading phase

After the calibration phase, participants were automatically directed to the reading phase.

During the reading process, the screen initially displayed the serial number of the current chapter.

Subsequently, the text appeared with three lines per page, ensuring each line contained **no more than ten Chinese characters (excluding punctuation).**

- *Need to control the length of text that we show to the participants*

On each page, the **middle** line was highlighted as the **focal point**, while the **upper and lower lines were displayed with reduced intensity as the background.** Each character in the middle line was sequentially **highlighted** with red color for 0.35 s, and participants were required to read the novel content following the highlighted cues.

- *May serve as a direction guide*

- It should be noted that during the initial participation in the experiment, participants were required to complete a **practice reading phase**.

The preface chapter of The Little Prince was selected as the reading material for this phase. All settings remained the same as those of the formal reading stage, to familiarize participants with the eye-tracker calibration process and the reading task. After each run, participants were provided with **adequate rest time** until they reported ready to start the subsequent run.

- During the rest period, the experimenter checked the power status of the eye-tracker and replaced the batteries as necessary to ensure its continuous operation.

Data collection and analysis

Eye-tracking data collection Eye-tracking data was acquired using Tobii Pro Glasses 3 (see Figure 1a). The device features 16 illuminators and 4 eye cameras integrated into scratch-resistant lenses, along with a wide-angle scene camera, allowing for a comprehensive capture of participant behavior and environmental context. We utilized the package g3pylib to control the glasses. During recording, the sampling rate was set to 100 Hz. The raw data was exported to .zip files.

- *Need to update according to our device*

Paper 2: Eye Movement Patterns in Natural Reading: A Comparison of Monolingual and Bilingual Reading of a Novel

Key Findings:

- Bilingual L2 reading patterns show longer sentence reading times (20%), more fixations (21%), shorter saccades (12%) and less word skipping (4.6%), than L1 reading patterns.
- These distinctive patterns suggest that eye-tracking metrics can reliably differentiate between reading in dominant versus non-dominant languages
- Proficiency Effects: The study found that L2 proficiency influenced fixation counts, with higher proficiency correlating with fewer fixations.
- Importantly, even highly proficient L2 readers maintained distinctive eye movement patterns compared to L1 reading.

Related Theory:

- The authors explain their findings through the E-Z reader model, suggesting that L2 reading differences stem primarily from slower lexical processing
- This theoretical framework can inform our interpretation of eye movement patterns around code-switching points, where lexical access might be particularly affected.

Link

Paper 3: What Can Eye Movements Tell Us about Reading in a Second Language: A Scoping Review of the Literature

Link

Data were analyzed on two levels: global (paragraph) and local (word). Results from the experiment showed significant differences in the global level on all measures except fixation duration, showing that bilinguals were slower to read, had more saccades, and had more fixations than monolinguals. Conversely, results from the local level showed significant differences in all measures except regressions out and skipping rate, showing that bilinguals had longer fixations, gaze duration, and total reading times.‘

Weaker Links Hypothesis (also known as the frequency-lag hypothesis), which hypothesises that bilingual individuals split their time between two languages, resulting in less experience with words in each language compared to monolinguals.

Variables measured: first fixation duration, single fixation duration, gaze duration, total reading time, and skipping probability.

One interesting result:

- bilinguals with more exposure to a second language have similar eye movement behavior to monolinguals

Linguistic distance: The similarity and difference between languages themselves can influence the eye movements

Should I stay or should I switch? A cost-benefit analysis of voluntary language switching in young and aging bilinguals

Grosjean, F. (2010). *Bilingual: Life and Reality*. Harvard University Press. http://www.jstor.org/stable/j.ctt13x0ft8

Compensatory effects of individual differences, language proficiency, and reading behavior: an eye-tracking study of second language reading assessment

1. **Fixation Duration and Language Proficiency**:

- Shorter **fixation durations** were generally predictive of higher comprehension scores across tasks, suggesting that more proficient readers (with higher L2 proficiency) tend to process text more efficiently.

- **Higher L2 proficiency** allowed participants to make quicker and more efficient fixations, as proficient readers spent less time on each word and were able to focus more on understanding the overall meaning of the text.

2. **Compensatory Effects**:

- The study found that **proficient readers** (high L2 proficiency) could compensate for **longer fixations** or slower processing by relying on other cognitive abilities, such as **reasoning** and **motivation**.

- In contrast, **lower proficiency readers** were more reliant on efficient reading strategies, such as **shorter fixation durations**, to achieve better comprehension. However, their ability to compensate with other cognitive skills was limited compared to proficient readers.

3. **Task-Specific Differences**:

- In the **summary task**, which required deeper processing of the text, participants with **higher L2 proficiency** were able to compensate for **longer fixation durations** through other strengths (e.g., motivation, strategic reading skills). This suggested that proficient readers may rely more on their cognitive resources to handle more demanding tasks.

- For simpler tasks like the **cloze** and **multiple-choice tasks**, proficient readers displayed greater efficiency, with **shorter fixations** correlating with higher scores.

Conclusion:

In summary, the study highlights that **L2 proficiency** influences **fixation behavior**: higher proficiency is associated with shorter and more efficient fixations, while lower proficiency readers rely more on fixation efficiency to compensate for slower processing. These findings suggest that proficient readers can use their language and cognitive skills more flexibly, compensating for less efficient fixation behavior during more complex reading tasks.

Paper 4: An eye-movement database of bilingual language control for Chinese-English bilinguals

- Link

The article titled "An eye-movement database of bilingual language control for Chinese-English bilinguals," published in *Scientific Data* on February 25, 2025, presents a comprehensive eye-movement dataset aimed at enhancing the understanding of bilingual language control mechanisms. Nature

Background: Bilingual individuals often experience simultaneous activation of both languages, leading to potential cross-language interference. Effective management of this interference, known as bilingual language control, is crucial for seamless communication. While previous studies have explored this phenomenon, there has been a lack of eye-tracking databases focusing on how different types of intra-sentential code-switching influence language control processes.Nature

Study Overview: To address this gap, the researchers compiled an eye-movement database by recording participants during a silent reading task that incorporated a language-switching paradigm. The study involved 80 Chinese-English bilinguals, divided into two groups based on their English proficiency:Nature

- **High-proficient group:** 40 participantsNature
- **Low-proficient group:** 40 participantsNature

Each participant read 1,280 sentences across various conditions:Nature

- Chinese-only sentencesNature
- English-only sentences
- Three types of intra-sentential code-switching sentencesNature

The sentences were structured with a subordinate clause preceding the main clause, with the subject of the main clause serving as the critical word where language switching occurred.Nature

Data Collected: The database includes eye-movement measures for 160 Chinese words and their English equivalents, capturing metrics such as:Nature

- First Fixation Duration (FFD): Reflects early lexical accessNature+1Nature+1
- Regression Path Duration (RPD): Sensitive to processing difficulties at later stagesNature

Applications: This dataset enables researchers to investigate:Nature

- The impact of different types of intra-sentential code-switching on bilingual language controlNature

- The influence of second language proficiency on language control mechanismsNature

By providing detailed eye-tracking data, the study offers valuable insights into the cognitive processes underlying bilingual language control, contributing to fields such as psycholinguistics and second language education.

Paper 5: PREDICTING LANGUAGE PROFICIENCY IN BILINGUAL CHILDREN

https://www.proquest.com/docview/2423130556?accountid=14749&parentSessionId=wetoQn18abkNOpiN6BZ5SRRK562EoDiCYhgt93FjVLA%3D&pq-origsite=primo&sourcetype=Scholarly%20Journals

Summary:

- Language dominance and bilingualism are continua, not fixed categories.
- Thresholds must be interpreted considering SES and cognitive abilities, which also significantly influence proficiency.
- Bilingual children often catch up to monolinguals by the end of primary school and may eventually outperform them in other areas.
- This study contributes a data-driven, nuanced model of bilingual development, acknowledging its complex, multi-dimensional nature.

Paper 6: Bilingual Language Processing and Interference in Bilinguals: Evidence From Eye Tracking and Picture Naming

https://onlinelibrary-wiley-com.libproxy2.usc.edu/doi/full/10.1111/j.1467-9922.2007.00401.x

Summary:

Experiment 1: Russian words caused more eye fixations and longer naming times than nonword or English translation controls, showing strong interference due to recognition of orthographic Russian words.

Experiment 2: Phonological Russian words also attracted more eye movements than nonwords, indicating recognition, but naming times were similar for both. This suggests both types of

distractors interfered with production for different reasons—phonological Russian words due to lexical competition and nonwords due to pronounceability.

Eye-tracking vs. Reaction-time: Eye-tracking data showed clear differentiation between Russian words and nonwords, but reaction times did not always reflect that—especially in Experiment 2. This implies that recognition of a nontarget word does not always interfere with production.

Proficiency and Expectation: Participants' higher proficiency and reading speed in English may have allowed them to process English distractors more efficiently and ignore them, while Russian words were more disruptive due to unexpectedness and lower reading fluency.

Processing Routes: The results align with dual-route reading models—Russian words (unfamiliar but real) and nonwords might both be processed through sublexical routes, though recognition patterns differed.

Eye movements are a stable indicator of word recognition in bilinguals.

Reaction-time delays are more influenced by the pronounceability of distractors rather than their lexical status.

Recognition of nontarget language occurs even without overt interference in production.

Differences in language dominance, expectations, and reading proficiency modulate crosslinguistic interference.

Further research is needed to clarify whether interference is due to direct lexical competition or activation of translation equivalents.

Paper 7: Eye Tracking in Second Language Acquisition and Bilingualism

https://www-taylorfrancis-com.libproxy2.usc.edu/books/mono/10.4324/9781315775616/eye-tracking-second-language-acquisition-bilingualism-aline-godfroid

It's a book but seems extremely relevant to the topic + data analysis particularly.

Paper 8: VISUAL WORD RECOGNITION IN BILINGUALS: EYE-TRACKING EVIDENCE THAT L2 PROFICIENCY IMPACTS ACCESS OF L1 PHONOTACTICS

https://www.proquest.com/docview/2691702230?accountid=14749&parentSessionId=9Sw%2BEUFnWLy1NBV4GeOEjBrSV5YjW6zx6JkxYlL0X9M%3D&pq-origsite=primo&sourcetype=Scholarly%20Journals

Summary:

This study explored how bilinguals with varying levels of second-language (L2) proficiency activate first-language (L1) phonotactic constraints—rules about sound combinations—during visual word recognition. Across two experiments, lower-proficiency bilinguals showed increased eye movements toward words that aligned with their L1 constraints, even without auditory cues. The results suggest that visual orthographic input alone can trigger phonological and phonotactic activation across languages. While no general differences were found between monolinguals and bilinguals, L2 proficiency influenced the degree of cross-linguistic interference, with higher-proficiency bilinguals better able to inhibit L1 influence. These findings have implications for models of bilingual language processing, indicating that both bottom-up (sensory-driven) and top-down (knowledge-driven) mechanisms shape language coactivation, even at sublexical levels, and that task difficulty and input modality further modulate this interaction.

Paper 9: Chinese as a Second Language Assessment

https://link-springer-com.libproxy2.usc.edu/book/10.1007/978-981-10-4089-4

Another relevant book

Paper 10: Cognate facilitation in Frisian–Dutch bilingual children's sentence reading: An eye-tracking study

https://www-sciencedirect-com.libproxy2.usc.edu/science/article/pii/S0022096519301407

Summary:

This study examined cognate facilitation in Frisian–Dutch bilingual children during sentence reading. The results showed a non-gradual cognate facilitation effect in Frisian—children read identical cognates faster than non-identical cognates and non-cognates—while no such effect appeared in Dutch. This suggests that children draw on their dominant reading language (Dutch) when reading in their non-dominant one (Frisian), but not the other way around. The findings align with prior research showing that cross-language activation depends on reading proficiency and are consistent with models like BIA+ that emphasize weaker lexical connections in the less dominant language. The study also found that cognate effects were more pronounced in later processing stages, possibly due to the overlap between Frisian and Dutch vocabularies. Limitations include the lack of counterbalancing and limited data for proficiency analysis. Importantly, the study highlights that reading fluency transfers across languages, supporting educational approaches that promote literacy development in both languages.

Paper 11: Ideal L2 Self, Visual Learning Styles, and L2 Self Confidence in predicting Language Proficiency and L2WTC: A Structural Equation Modeling

https://www.proquest.com/docview/2932509928?accountid=14749&parentSessionId=9zdw5qyEip85ZjPLl9A214VGjY6BrdxsE1svs0JdeSQ%3D&pq-origsite=primo&sourcetype=Scholarly%20Journals

Summary:

This study highlights the important role of visual learning style, Ideal L2 self, self-confidence, and language proficiency in shaping learners' willingness to communicate (L2WTC) in a second language. Visually oriented learners demonstrated stronger Ideal L2 selves through imaginative capacity, which, in turn, fostered greater self-confidence and communication willingness. The findings support previous research and underscore that motivated and proficient learners are more likely to engage in L2 communication. Implications for language educators include the importance of helping learners visualize their future L2 selves, identify the gap between their current and ideal abilities, and build confidence. However, limitations such as reliance on self-reported data, lack of gender control, a focus only on the Ideal L2 self, and a homogenous proficiency level suggest caution in generalizing results. Future research should address these gaps by including diverse learning styles, broader proficiency levels, and additional components of the L2 Motivational Self System.

Paper 12: Predicting First-Language and Second-Language Proficiency Using Eye Fixation Data and Demographic Information: Assumptions, Data Representations, and Methods

https://doaj.org/article/544f543cdbab41b3874de1e1cf022b03

Summary:

This study explores the use of eye fixation data and demographic information to predict first language (L1) and second-language proficiency (L2P) through artificial intelligence (AI) methods. Analyzing the MECO L2 dataset, the findings reveal that L1 can be moderately predicted using only eye fixation data, though L2P prediction is limited when relying on this data alone. Among the L2P assessments, the Lexical Skills battery proved most effective. However, combining eye fixation with demographic data substantially improved prediction performance for both L1 and L2P, achieving high accuracy and diminishing the relative contribution of eye fixation features alone. Additionally, 2D-scatter plot images emerged as a promising representation format for AI training using eye fixation data, particularly in L1 prediction.

Paper 13: Application of Computational Linguistics to Predicting Language Proficiency Level of Persian Learners' Textbooks

https://www.proquest.com/docview/2703422619?accountid=14749&parentSessionId=5XpP7wd%2FQM%2Baso3tInoCMDBRWcc%2F5H03E4VLe0gFQrc%3D&pq-origsite=primo&sourcetype=Scholarly%20Journals

Summary:

This study presents a computational linguistic model aimed at predicting language proficiency levels using various feature sets. To evaluate performance, classifiers were trained on different combinations of statistical and linguistic features, with Term Frequency-Inverse Document Frequency (TF-IDF) serving as a baseline. Seven distinct learning scenarios were designed, ranging from combined feature sets to isolated linguistic dimensions like phonetics, morphology, syntax, and semantics.

Results, as summarized in Table 4 (confusion matrix), show that classifiers performed poorly with TF-IDF alone—misclassifying all texts. However, combining statistical and linguistic features significantly improved accuracy, particularly for intermediate levels. Each feature set influenced classifier behavior differently, demonstrating the nuanced role of linguistic properties in proficiency prediction. The model's effectiveness emphasizes the benefit of rich linguistic features over basic frequency-based methods in accurately assessing language levels.

Paper 14: Incidental Vocabulary Learning From Bilingual Subtitled Viewing: An Eye-Tracking Study

https://onlinelibrary-wiley-com.libproxy2.usc.edu/doi/full/10.1111/lang.12495

Summary:

This study investigated the effects of different subtitle types—captions, bilingual subtitles, L1 subtitles, and no subtitles—on incidental vocabulary learning and word processing among intermediate to advanced Chinese learners of English. Results showed that all subtitle types supported vocabulary learning, with form recognition being easiest, followed by meaning recognition and meaning recall. Captions were most effective for learning word forms, while bilingual subtitles were most beneficial for meaning recognition and outperformed L1 subtitles in meaning recall. Eye-tracking data revealed that participants using bilingual subtitles focused more on L1 translations, suggesting attempts to build form-meaning links, though time spent on L1 did not predict vocabulary gains. Instead, time spent on L2 words was a stronger predictor of learning outcomes. The study highlights the value of bilingual subtitles for meaning acquisition but points to the need for additional data to understand underlying cognitive processes and recommends further research with varied learner profiles, content types, and delayed testing.

Paper 15: The Interaction of Linguistic and Visual Cues for the Processing of Case in Russian by Russian-German Bilinguals: An Eye Tracking Study

https://onlinelibrary-wiley-com.libproxy2.usc.edu/doi/full/10.1111/tops.12724

Paper 16: Prediction at the Discourse Level in Spanish–English Bilinguals: An Eye-Tracking Study

https://doaj.org/article/01f3b83c02914e7ca2eb3c5a18cdbd04

Paper 17: Lexico-syntactic interactions during the processing of temporally ambiguous L2 relative clauses: An eye-tracking study with intermediate and advanced Portuguese-English bilinguals

https://go-gale-com.libproxy2.usc.edu/ps/i.do?p=SCIC&u=usocal_main&id=GALE%7CA587088523&v=2.1&it=r&sid=summon

Paper 18: Bilingual Language Processing and Interference in Bilinguals: Evidence From Eye Tracking and Picture Naming

https://onlinelibrary-wiley-com.libproxy2.usc.edu/doi/full/10.1111/j.1467-9922.2007.00401.x

Paper 19: Designing Research on Bilingual Development

https://link-springer-com.libproxy2.usc.edu/book/10.1007/978-3-319-11529-0

[An eye-movement database of bilingual language control for Chinese-English bilinguals | Scientific Data]

https://osf.io/8j9uv/files/osfstorage

(insert lit review of dataset we're using)

1. Data Inspection and Cleaning
Missing Values: Check for missing values in your dataset and decide on a strategy to handle them. Common methods include mean/median imputation or interpolation.
Outliers: Identify and address outliers using statistical methods (e.g., Z-score, IQR) to ensure they do not skew your analysis.

Important Features:
Regression Path Duration (Column 11) is especially valuable as it's "sensitive to the detection of processing difficulty at later stages of sentence processing". This measure captures the total time spent on regression paths for each word, which can reveal how bilinguals handle the cognitive load when switching between languages.

First Fixation Duration (Column 8) is crucial because it "can reflect early lexical access", making it a primary indicator of initial word processing differences between languages in code-switching contexts.

Gaze Duration/First Run Dwell Time (Column 10) provides insight into the total time spent on first-pass reading, which has been used in studies to demonstrate the code-switching effect.

IA Skip (Column 15) and Fixation Count (Column 14) are important as they provide spatial dimensions of eye movements, showing which words receive attention and which are skipped during code-switching.

Regression In/Out Counts (Columns 17-18) are valuable for tracking how often readers need to revisit words when switching languages, indicating processing difficulties.

2. Data Transformation and Normalization
Normalization: Scale your data to a common range using techniques like Min-Max Scaling or Standardization (Z-scoring). This is crucial for many machine learning algorithms to ensure features are on the same scale.

Feature Engineering: Consider creating new features that might be relevant for your analysis. For example, you could calculate the average fixation duration or the total dwell time across different areas of interest.

3. Feature Selection
Correlation Analysis: Perform a correlation analysis to identify highly correlated features. This can help in reducing dimensionality by removing redundant features.

4. Data Split
- Train-Test Split: Divide your dataset into training and testing sets (e.g., 80-20 split) to evaluate model performance accurately.
- Recommended Split: 30-70?

https://www.realeye.io/blog/post/eye-tracking-data-analysis

https://github.com/adolphslab/adolphslab_eyetracking

Paper 1: Using machine learning to detect events in eye-tracking data

Summary: The study uses Random Forests (RF) to classify speakers into monolingual or multilingual categories based on verbal fluency data.

Method:

- Used RF for classification and feature importance ranking
- Compared RF performance with other machine learning methods (SVM, kNN)
- Applied cross-validation for model evaluation

Results:

- RF outperformed other methods in classification accuracy
- RF provided interpretable results through feature importance rankings
- The model achieved high accuracy in distinguishing between monolingual and multilingual speakers

Why Relevant

- Demonstrates successful application of RF in language-related classification tasks
- Shows RF's effectiveness in handling complex, multidimensional linguistic data
- Highlights RF's ability to provide interpretable results, which could be valuable for understanding eye-tracking features in code-switching

Implications: We can possibly use Random Forests machine learning technique as suggested by Steve to develop our model to predict bilingual code switching.

From AI regarding how to ensure reliability and validity:

1. Implement Rigorous Cross-Validation
    - Use stratified k-fold cross-validation for your eye-tracking data
    - Consider leave-one-subject-out validation to test generalizability across participants
    - Report confidence intervals alongside performance metrics
2. Conduct Algorithm Comparisons
    - Compare RF with other algorithms suitable for time-series data (LSTM, XGBoost)
    - Justify RF selection based on empirical performance and interpretability needs
    - Use consistent evaluation metrics across all models
3. Leverage Feature Importance for Insight
    - Extract feature importance rankings to identify key eye-tracking predictors of code-switching
    - Connect these important features to psycholinguistic theories
    - Use permutation importance to validate feature rankings
4. Document Parameter Selection Process
    - Systematically test RF parameters relevant to eye-tracking data
    - Report optimal settings and their justification
    - Consider temporal dependencies in your eye-tracking data when configuring the model
5. Address Eye-Tracking Specific Challenges

- Implement preprocessing steps specific to eye-tracking (fixation identification, noise reduction)
- Consider temporal windows before code-switching events
- Account for individual differences in baseline eye movements

Model Selection and Hyperparameter Tuning

- Model Choice: **Random Forest** or **Support Vector Machines**

Recursive Feature Elimination: Start with all features, then iteratively remove the least important feature and retrain the classifier until performance begins to degrade significantly. The paper used a combined importance metric from three measures: Mean Decrease Impurity (MDI), Mean Decrease Accuracy (MDA), Univariate Feature Selection (UFS)

Track fluency in the model – eye fixation

Model: **Support Vector Machines (SVM)**

https://link.springer.com/article/10.3758/s13428-017-0860-3#Sec2

https://www.frontiersin.org/journals/neurorobotics/articles/10.3389/fnbot.2021.796895/full

1. Eye movement patterns in Natural Reading: A comparison of monolingual and bilingual reading of a novel
2. Bilingual: Life and Reality
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21. Bilingual Language Processing and Interference in Bilinguals: Evidence From Eye Tracking and Picture Naming
22. Designing Research on Bilingual Development

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Eye Movement Patterns in Natural Reading: A Comparison of Monolingual and Bilingual Reading of a Novel

Monolingual Eye Movements while Reading

When we read, our eyes move from one position to the next in order to identify and process visual word form information. This entails rapid jerk-like movements (saccades) and short periods of steadiness (fixations). Saccades are necessary to direct the gaze to a new location, bringing new information into the center of the visual field where acuity is best. During these saccades, no meaningful new visual information is gathered. They occur several times per second and typically move the eyes forward about 7-9 character spaces (for reviews:[16, 17]. Psycholinguists assume that eye movements during reading reflect language processing [18], with fixation durations as a marker of the ease of accessing the meaning of a word and integrating this into the current sentence. Because of the spatially accuracy and high temporal resolution of eye tracking, it allows us to dissociate early from late eye movement measures. In combination with other information, such as word length/frequency, this makes it possible to investigate the time course of the reading process. Additionally, reading processes in eye tracking are not confounded by task-related processes or strategies that other lab tasks (e.g. lexical decision or naming) entail. Hence, this method is considered to be the closest experimental parallel to the natural reading process.

During the last three decades, the development of monolingual theories on visual language comprehension has been heavily influenced by eye tracking research in reading. Rayner's influential review article [16], now 15 years old, already discusses more than 550 articles investigating this topic (for a more recent review:[17]). Also, several corpus studies of eye movements were undertaken, and these data were used to provide an account of (monolingual) reading. The Potsdam Corpus [19, 20] contained eye movements of 222 subjects reading 144 constructed German sentences (1 138 words). The Dundee corpus [21], an English and French study in which 10 participants read 50 000 words in paragraphs, was used to investigate effects of parafoveal processing. Clearly, these corpora of eye movements provide a very rich and extended source of information about the mechanisms that underlie language processing in a more natural context and could serve as harvesting grounds for the development of comprehensive language models. For example, the Amherst Sentence Corpus [22] was used to develop the first version of the SWIFT model of saccade generation [23].

The E-Z reader model [22, 24-29] is the most cited model of monolingual eye movements. It is implicitly limited to native language or even monolingual reading behavior, and it is yet unknown how these mechanisms operate when bilinguals read in

a second language, or how knowledge of a second language influences native language reading. However, it is interesting that the original E-Z reader model has been successfully accommodated to account for other reading patterns, such as those of older readers [30], children[31], or of non-alphabetic languages[32]. This illustrates that this model could be useful and relevant in future modeling efforts concerning bilingual eye movement patterns, and we will therefore align our analyses of bilingual reading behavior with the core assumptions and variables of this model.

The E-Z reader model assumes serial lexical processing. The completion of an early stage of lexical processing on word *n* , called the *familiarity check* , is the 'trigger' that causes the oculo-motor system to begin the programming of a saccade directed towards the next word *n+1* . The subsequent completion of a second stage of lexical processing on word *n* , called the *completion of lexical access* , causes attention to shift from word *n* to word *n+1* . Thus, the programming of saccades is decoupled from the shifting of attention, which is allocated serially to only a single word at a time [22]. Because attention shifts are faster than the programming of a saccade[30], the lexical processing of word *n+1* usually begins when the eyes are still fixated on word *n* . This feature of the model allows parafoveal processing of upcoming words. Following similar reasoning, the model predicts that parafoveal words, which are processed fast enough, might be skipped.

The model assumes that word length and frequency are important lexical variables that have a large effect on the eye movements, because these variables define the duration of the familiarity check [33, 34]. Consequently, they determine fixation duration, fixation count, rightward saccade length, skipping and regression rates. These will also be the core variables that will be assessed in the present paper.

Theories about Bilingual Word Recognition

The most cited, and the only implemented, model of bilingual visual word recognition is the Bilingual Interactive Activation plus (BIA+) model [36]. This model is an adaptation of the interactive activation model of word recognition [63]. The main differences are the inclusion of lexical representations of two languages, and a distinction between a word identification system and a task/decision system. The BIA+ states that during bilingual reading there is parallel, language independent activation of lexical representations in an integrated lexicon. Language nodes that represent language membership are included in the model, but they cannot tune word recognition towards a single language via top-down activation. This architecture implies that for every word bilingual readers encounter all lexical candidates from all known languages are activated to some extent. Evidence for this model is generated by studies supporting cross-lingual interactions (see previous paragraph for references).

A limitation of the BIA+ model [36], similar to the monolingual interactive activation model [63] is that it is tailored to isolated word recognition, and not to sentence reading. The authors do assume effects of sentence context and non-linguistic information on word recognition but the exact nature of these interactions are not specified. This means that a model of bilingual eye movements, such as the E-Z reader model, is not yet available, as there is also no sentence reading data to base it upon.

The *weaker links account* [13, 64], sparked by small but consistent production disadvantages exhibited by bilinguals compared to monolinguals [65-67, 13, 68], has recently gained popularity in the literature. Like the BIA+ model [36], it assumes an integrated bilingual lexicon. According to this *frequency-lag* account, bilinguals will have about double the amount of lexical items in their lexicon as monolinguals and will necessarily divide the frequency of use of these words between languages [64]. Considering the lexical quality hypothesis [69,70], which states that increased word practice results in better precision of the corresponding lexical representations, it is plausible that bilingual representations will be of lower precision than those of monolinguals. Indeed, Gollan, Montoya, Cera and Sandoval predict that weaker links between word form and representations for bilinguals should result in slower lexical access during language comprehension, either while accessing L1 or L2, compared to monolinguals [64]. Effects might be smaller than in production because the processes needed for language production are less practiced, more difficult and involve more levels of processing for which frequency is important [13]. In the comprehension domain, it was indeed found that bilinguals show slower L1 lexical decision times than monolinguals do [71, 72].

A core assumption at the heart of the weaker links account is that total language exposure is equal for all people. While this maybe the case for bilinguals who are exposed to two languages from birth, it is definitely not true for all groups of bilinguals. The authors that constructed the weaker links account used mostly early Spanish English bilinguals [13, 64, 67]. A population of unbalanced bilinguals usually acquires a 2nd language in a classroom context, thus increasing their total vocabulary and language exposure, not per se decreasing their L1 exposure. On top of that, the words of their mother tongue will have been fully lexically entrenched before they start learning their second language. This means that for late learners of an L2, the lexical entrenchment for L1 words might be equally strong as the lexical entrenchment for the words of a monolingual.

Predictions Monolingual vs. Bilingual Reading

For bilinguals, reading experience is supposedly spread across two different languages, L1 and L2 [64]. This implies lower absolute exposure to each language, which could

result in slower lexical access and thus word recognition [71, 72] and reading for bilinguals compared to monolinguals. We expect that the weaker links account does not apply to late bilinguals, per se, because these participants might have experienced larger language exposure in general than monolinguals have and because lexical entrenchment of L1 words is in an advanced stage before learning an L2.

Although Gollan et al.'s eye tracking study [13] does explicitly compare English monolinguals with balanced Spanish-English bilinguals on an English reading task, their bilingual group scored worse on the objective English proficiency measure than their monolingual group did [13]. Bilinguals accordingly showed longer gaze duration and lower skipping rates for the target words than monolinguals did. It is thus unclear whether this difference is a necessary and intrinsic consequence of bilingualism or rather whether it is driven by proficiency.

In our study, we excluded language proficiency as a possible confounding variable by matching our bilingual's L1 proficiency to our monolingual's language proficiency. Note, that similar proficiency scores would already imply that the lexical entrenchment of the bilinguals' L1 is on the same level as the lexical entrenchment of the monolinguals.

In conclusion, the weaker links account predicts slower sentence reading times, more and longer fixations per sentence, a smaller saccade length, lower skipping rates and higher regression rates, for bilinguals reading in L1 than for monolinguals. These differences will be subtler than the differences between the bilingual L1 and bilingual L2 reading pattern, because the L1 proficiency is the same for both groups. When we assume similar L1 lexical entrenchment for unbalanced bilinguals, we would expect a similar global eye movement pattern for monolinguals and bilinguals reading in their L1.

Model Fitting

For analysis, we selected the dependent variables that are well captured by models of reading such as the E-Z reader model. For both comparisons, the dependent measures under investigation are: a) sentence reading time including fixations and re-fixations, b) total number of fixations that landed in one sentence, c) the average fixation duration of the fixations that landed in that sentence, d) the average rightward saccade length per sentence, e) the probability of making an inter-word regression towards or within a certain sentence and f) the probability of first pass skipping.

Our data corpus was analyzed with linear mixed effects models with the lme4 (version 1.1-7) and lmertest (version 2.0-20) package of R (version 3.0.2) [85], because a multilevel design is the best way to statistically control for a range of predictors that in this experiment we could not or did not want to manipulate.

For the first within-subject comparison of the bilingual L1 vs. L2 reading data, the same fixed effects model was fitted for every eye movement measure. The fixed factors were language (L1 or L2), number of words per sentence (continuous), average word length per sentence (continuous), average frequency of the content words per sentence (continuous) and L2 proficiency (continuous). This last variable is the composite proficiency score presented in Table 1. Note that this variable represents something different for the two language conditions. For the L2 condition this is the language they are reading in. For the L1 condition it is their proficiency in a second language that they do not use in this condition. For the content word frequency, the subtitle word frequency measures[86, 87] of the content words in a particular sentence were log transformed to normalize their distribution. All continuous predictors were centered. The absolute value of the maximum correlations among main effects was under 0.51 for all eye movement measures (<0.506 for Saccade length, <0.156 for fixation count, <0.167 for fixation duration, <0.249 for dwell time, <0.386 for regressions, <0.245 for skips).

In a first step, we fitted a "complete" model. The fixed part of the model contained all main effects and interactions (up to 5-way) and the random part contained two random clusters: one for subject (the participant ID-number) and one for sentence (the sentence ID-number). After fitting this first model, we excluded the terms one by one, starting with the factor that contributed the least to the fit. By model comparisons, we decided when we arrived at the best possible fit. Then we added random slopes one by one. When they contributed to the fit, we included the slope in the model. We choose to test addition of every possible random slope, and strive for a maximal random structure [88]. We added, in this order, language as a random slope for each sentence and language, word length, word frequency and number of words as random slope per subject. For the count variable and the binomial variables (fixation count, skipping rate, regression rate) we report the p-values for the significant effects. For the continuous variables (sentence reading times, average fixation duration and saccade length), we obtained the p-values by computing the F-Test with Kenward-Roger adjusted degrees of freedom [89] for our fixed effects in the final models.

For the second important (between-subject) comparison between the bilingual L1 and monolingual L1 reading, the same model was fitted for every eye movement measure. Here, the fixed factors were bilingualism (Bilingual or Monolingual), number of words (continuous), average word length (continuous), average frequency of the content words (continuous) and L1 proficiency (continuous). This last variable is the composite proficiency score presented in Table 1. Note that for both the bilinguals and the monolinguals this is the language they are reading in. The frequency measure was computed the same way as in the previous comparison. The process of top-down fitting of fixed effects and bottom-up fitting of the random slopes was identical to the process in the first comparison. Again, a maximum random structure was aspired but this time

we added, in this order, bilingualism as a random slope for each sentence and word length, word frequency and number of words as random slope per subject. Again, the p-values for the continuous variables were calculated with the F-test with Kenward-Rogers adjusted degrees of freedom [89].

What Can Eye Movements Tell Us about Reading in a Second Language: A Scoping Review of the Literature.

Our findings have the potential to improve empirical approaches used to explore bilingual reading and highlight the importance of several factors when designing such experiments and interpreting their results. Characteristics of a bilingual's first language will affect their L2 reading in different ways. For example, if the orthographies of L1 and L2 are similar, L2 reading may be less affected compared to if the L1 and L2 orthographies are more different. Therefore, understanding the characteristics of L1 and the similarities and differences between one's L1 and L2 can inform instructional practices. It has been established that cross-linguistic transfer facilitates L2 reading acquisition by allowing the reader to recognize L1 features during L2 reading, which ultimately enhances their conceptual understanding and skill application for both languages. Recognizing which factors come to bear on L2 reading acquisition is crucial for improving reading programs for bilinguals. Moreover, findings of these studies support that improved L2 proficiency positively impacts L2 reading proficiency, implying the need to focus instruction on improving oral proficiency in order to improve reading proficiency. It should also be noted that, although highly proficient L2 readers approximated monolingual readers' eye movement performance, this effect was modulated by the characteristics of the L1 orthography. This indicates that bilinguals whose L1 and L2 are relatively more similar are, in turn, more prone to facilitatory effects. Our findings emphasize the importance of considering L2 proficiency and L1 and L2 characteristics (beyond merely considering transparency) when designing and conducting investigations of bilingual reading processes.

Background:
Bilingual language processing has been widely studied, revealing distinct eye movement patterns in bilinguals reading in their second language (L2). Bilinguals typically show longer reading times, more fixations, and shorter saccades compared to monolinguals, reflecting increased cognitive load (Mou et al., 2024). The Weaker Links Hypothesis suggests this occurs due to divided cognitive resources between languages (Cop et al., 2015), though higher L2 proficiency can reduce these effects (Quiñonez-Beltran et al., 2024). Additionally, language switching incurs cognitive costs, with more pronounced switching behavior among frequent code-switchers (Gollan & Ferreira, 2009). Eye-tracking technology provides valuable metrics, such as gaze duration and fixation count, which can reveal these cognitive processes. This research aims to explore how these eye movement patterns, particularly in code-switched sentences, can predict bilingual language proficiency, offering a deeper understanding of bilingual cognitive control during reading.

This study explores how machine learning (ML) analysis of eye-tracking data can reveal the relationship between code-switching and visual attention in bilinguals. Code-switching, the switching of languages during communication, offers insight into the cognitive mechanisms underlying bilingual processing. By analyzing how bilinguals process code-switched Chinese-English sentences, our research uses a Random Forests classification model to predict second language proficiency based on eye movement. This approach combines eye-tracking measurements with pattern recognition to identify how attention patterns during language switching correlate with proficiency levels. Our interdisciplinary methodology merges cognitive science, linguistics, and computational modeling to reshape how we understand bilingual language control. Predicting proficiency using real-time data enables more efficient language assessment tools for educational and clinical settings. Insights into cognitive load during language-switching can inform personalized teaching strategies, while this research also advances AI systems for bilingual language use, improving machine translation, natural language processing, and human-computer interactions. By examining the relationship between visual attention and language proficiency, the study aims to uncover how humans navigate multiple language systems, and has far-reaching implications in an increasingly multilingual world.

Eye-tracking technology provides valuable metrics, such as gaze duration and fixation count, which can reveal these cognitive processes. This research aims to explore how these eye movement patterns, particularly in code-switched sentences, can predict bilingual language proficiency, offering a deeper understanding of bilingual cognitive control during reading.

Studying bilingual language processing has revealed distinct eye movement patterns in bilinguals reading their second language (L2), including longer reading times, more fixations, and shorter saccades compared to monolinguals, all of which reflect increased cognitive load (Mou et al., 2024). The *Weaker Links Hypothesis* suggests this occurs due to divided cognitive resources between languages (Cop et al., 2015). Language switching also incurs cognitive costs, with more pronounced switching behavior among frequent code-switchers (Gollan & Ferreira, 2009). Eye-tracking technology provides valuable metrics, such as gaze duration and fixation count, which can reveal these cognitive processes. This study explores how machine learning (ML) analysis of eye-tracking data can reveal the relationship between code-switching and visual attention in bilinguals, and ultimately, bilingual language proficiency.